

SEIZURES/FITS

Give urgent attention to the patient who is unconscious and fitting:

- If current head injury ≥ 18 .
- Place in left lateral lying (recovery) position and give 100% face mask oxygen.
- Establish IV access.
- If glucose < 3 or unable to measure, give **dextrose 10%¹** 5mL/kg IV. If known alcohol user, give **thiamine** 100mg IM/IV before dextrose. Recheck glucose after 15 minutes: if still < 3 , give further **dextrose 10%¹** 2mL/kg IV. Once glucose ≥ 3 , continue **dextrose 5%** 1L 6 hourly.
- If ≥ 20 weeks pregnant up to 1 week postpartum $\rightarrow 159$.
- If not pregnant or < 20 weeks pregnant, give **diazepam** 10mg IV over at least 2 minutes or **midazolam** 10mg IM/buccal². If still fitting after 5 minutes, repeat diazepam/midazolam dose.
- If still fitting 5 minutes after second dose of diazepam/midazolam or patient does not recover consciousness between fits, refer urgently. If available, doctor to give **phenytoin** 20mg/kg IV in 200mL **sodium chloride 0.9%** (not dextrose) in a different line to diazepam, over 60 minutes with BP and ECG monitoring. If dysrhythmia develops, interrupt infusion and restart slowly. Refer urgently.

Approach to the patient who is not fitting now

Confirm that patient indeed had a fit: jerking movements of part of or the whole body, usually lasting < 3 minutes. May have had tongue biting, incontinence, post-fit drowsiness and confusion.

Yes

Refer patient same day if any of:

- Temperature $\geq 38^{\circ}\text{C}$, headache, neck stiffness or purple/red rash, **meningitis** likely: give **ceftriaxone** 2g IV³/IM. Avoid injecting $> 1\text{g}$ IM at one injection site.
- If patient was in malaria area and malaria test⁴ positive, also give **artesunate** 2.4mg/kg IM. If artesunate unavailable, give quinine as slow IV infusion over 4 hours: dilute **quinine** 20mg/kg in **dextrose 5%** 5-10mL/kg. If IV not possible, give IM⁵ diluted in **sodium chloride 0.9%**.
- New/different headache or headache getting worse/more frequent
- Patient with HIV and no known epilepsy
- Decreased consciousness > 1 hour after fit
- Glucose < 4 one hour after treatment or patient on glimepiride/insulin
- Glucose $\geq 11.1 \rightarrow 17$
- New sudden asymmetric weakness or numbness, difficulty speaking or visual disturbance
- BP $\geq 180/130$ more than 1 hour after fit has stopped
- Alcohol/drug use: overdose or withdrawal
- Recent head injury
- Pregnant or up to 1 week postpartum. If ≥ 20 weeks pregnant and just had fit $\rightarrow 159$.

No

New sudden asymmetric weakness or numbness of face, arm or leg; difficulty speaking or visual disturbance

Stroke or TIA likely $\rightarrow 136$.

Collapse with twitching lasting < 15 seconds following flushing, dizziness, nausea, sweating and with rapid recovery

Common faint likely $\rightarrow 28$.

If diagnosis uncertain, refer.

Approach to the patient who had a fit but does not need same day referral

Is the patient known with epilepsy?

Yes

Give routine **epilepsy** care $\rightarrow 149$.

No

- Doctor to check full blood count, creatinine (eGFR), urea, sodium, calcium and review results.
- If focal seizures or new fits after meningitis, stroke or head injury, discuss with specialist.
- If patient had ≥ 2 definite fits with no identifiable cause, doctor to consider epilepsy and give routine care $\rightarrow 149$.

¹If dextrose 10% unavailable: mix 1 part **dextrose 50%** to 4 parts water for injection to make dextrose 10% solution. ²Buccal: use 5mL syringe to draw up correct dose, remove needle and give midazolam between the cheek and gum. ³Do not mix Ringer's lactate and IV ceftriaxone. Flush IV line with **sodium chloride 0.9%** before and after IV ceftriaxone. ⁴Test for malaria with rapid diagnostic test if available, and parasite slide microscopy. ⁵To give IM quinine: first calculate volume of sodium chloride 0.9% in mL: weight $\times 20 \div 100$. Then add this volume of sodium chloride 0.9% to quinine 20mg/kg and inject half the volume into each thigh.